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A polyp from nothing: The extreme regeneration capacity of the Atlantic invasive sun corals *Tubastraea coccinea* and *T. tagusensis* (Anthozoa, Scleractinia)

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ABSTRACT

The azooxanthellate sun corals *Tubastraea coccinea* and *T. tagusensis* have invaded and dramatically changed rocky reef habitats along the Brazilian coast. Management of invaded areas has brought discussions on its potential negative effects regarding species dispersal, including those related to fragmentation. Although fragmentation has been observed in unmanaged localities, here we experimentally tested the sun coral ability to regenerate from undifferentiated coral tissue to fully functional polyps in response to differences in food supply, temperature, and fragment size. In general, survival and successful regeneration were high and markedly similar for both species. However, while percent of tissue retraction/reorganization (a proxy of metabolic cost) responded to fragment size and temperature, mouth development was faster in fed fragments. Together, our findings indicate that sun coral fragments may regenerate at temperatures that lead to extensive bleaching in native scleractinians that harbor *Symbiodinium*. Although no doubts remain about the necessity and efficacy of the management of invasive *Tubastraea* spp., recovery of undifferentiated coral tissue to fully functional polyps may contribute to their invasion success and should be taken into account during management efforts.

1. Introduction

The regeneration capacity of cnidarians is remarkable and plays an essential role in the maintenance of coral reef ecosystems. Most cnidarians repair tissue remnants to recover from physical damage, such as predation and wave impact (Cameron and Edmunds, 2014; Forsman et al., 2015; Lirman, 2000). In a few species, however, cell totipotency or "transdifferentiation", may allow regeneration into a complete animal from small fragments (Bosch, 2007; Holstein et al., 2003; Kramarsky-Winter and Loya, 1996; Piraino et al., 1996). Consequently, fragmentation combined with regeneration may contribute to small scale dispersal (Heyward and Collins, 1985; Highsmith, 1982; Jackson, 1986; Lirman, 2000; Smith and Hughes, 1999).

Nonetheless, regeneration processes are species-specific and triggered under certain environmental and biological conditions (Bak and Steward-Van Es, 1980; Bossert et al., 2013; Hall et al., 2015; Highsmith, 1982; Lea-Anne et al., 2005; Meesters and Bak, 1993; Sabine et al., 2015), such as temperature (Kramarsky-Winter and Loya, 2000) and colony health state (Ruiz-Diaz et al., 2016). Under such circumstances, coral species can exhibit higher regeneration rates and consequently invasive success, factors that need to be taken into account during attempts to manage their spread (Casado et al., 2014; Kramarsky-Winter and Loya, 1996).

Tubastraea coccinea Lesson, 1829 and *T. tagusensis* Wells, 1982 are invasive scleractinians that have dramatically changed shallow rocky reef species assemblages in the Southwestern Atlantic, threatening several endemic species (Creed, 2006; Lin et al., 2011; Luz and Kitahara, 2017; Mantelatto et al., 2011; Miranda et al., 2016; Santos et al., 2013). In addition to their diverse reproductive strategies and rapid settlement and growth (Vermeij, 2005; Glynn et al., 2008; Lages et al., 2011; Costa et al., 2014; de Paula et al., 2014; Capel et al., 2014), *Tubastraea* colonies have been observed recovering from fragmentation

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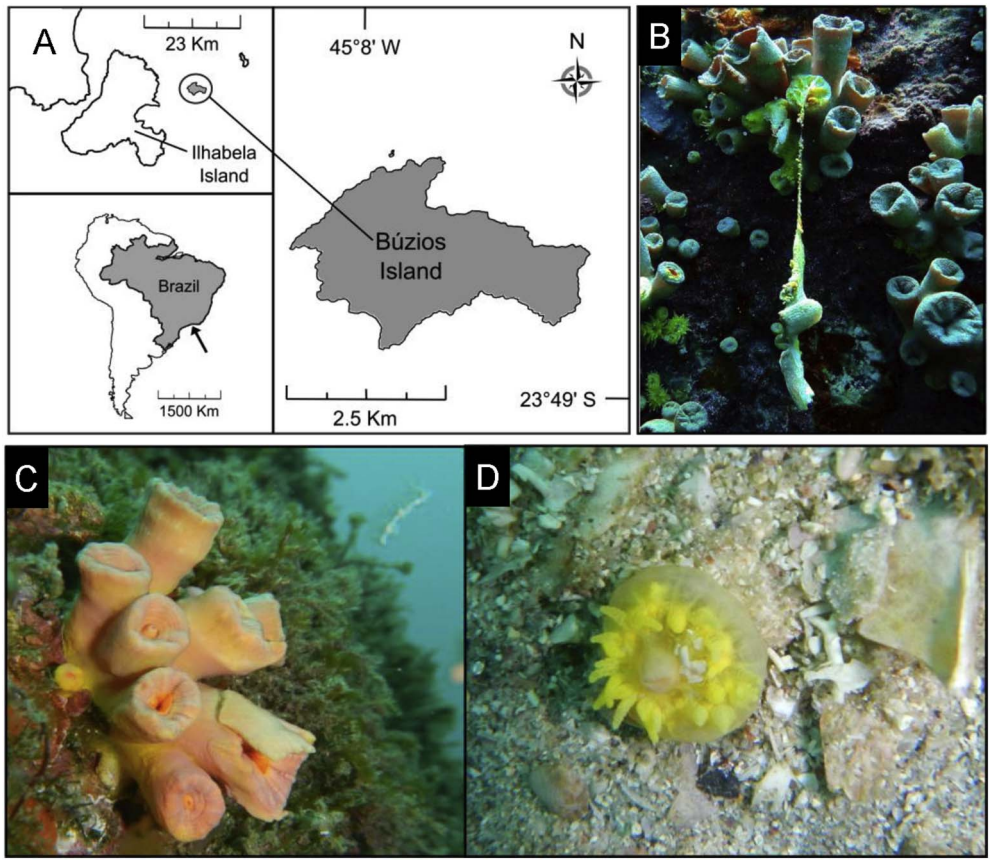

Fig. 1. A) Study area; B) A "naturally" injured *T. tagusensis* colony with broken fragments attached to mesenteries; C) *T. tagusensis* recovering from injury; D) A new viable polyp, possibly regenerated from a detached fragment, thriving on a sandy substrate.

and small polyps were found thriving on sandy substrates just below these colonies (Fig. 1). Such polyps likely originated from the regeneration of residual tissue from detached fragments, as previously reported for fungiid corals (Kramarsky-Winter and Loya, 1996; Vizel et al., 2011).

In order to test the regeneration capacity of *T. coccinea* and *T. tagusensis*, polyps were broken into two size classes and their regeneration ability tested under different conditions of temperature and food supply. As regenerative capacities probably differ among genetically distinct genotypes (see Meesters et al., 1996), in order to quantify only environmental responses of the phenomenon (i.e. excluding genetic variation), only one colony from each species were used herein. Colonies were kept in aquarium for 10 days in order to minimize previous environmental effects. Results show that both species have high regeneration capacity under diverse environmental conditions. Thus, such capacity needs to be taken in consideration during management efforts of these invasive species.

2. Material and methods

2.1. Laboratory procedures

One colony of each *T. coccinea* and *T. tagusensis* were collected at Búzios Island (Fig. 1a) and manually broken in laboratory to obtain, for each species, 120 fragments (skeleton with tissue) devoid of mesenteries, mouth or tentacles. For each species, fragments were then divided into two groups of 60 specimens based on size: 3.5–10.8 and 11.3–30.9 mm^2 for *T. coccinea*, and 3.8–11.3 and 13.2–53.5 mm^2 for *T. tagusensis*. Fragments were placed separately into 50 ml containers filled with 20 μm filtered seawater. For each combination of species and fragment size, individuals were further separated into three groups of 20 fragments, maintained at constant temperature of either 24 °C (historical average surface water temperature in the region), 27 °C (average summer sea surface temperature) or 30 °C (as observed during heat waves), in chambers set at constant photoperiod (12:12 h). Finally, effects of the presence of food were tested by adding an equal amount (10 ml) of live zooplankton (collected using a 150 μm mesh plankton net dragged for 5 min) every other day to half of the containers ($n = 10$) of each combination of remaining factors. Remaining zooplankton was removed from the containers the following day during water change. To avoid potential effects of fluctuating environmental conditions within chambers, every week all containers were changed between chambers (set to alternate experimental temperatures accordingly).

Water changes and fragment observations were carried out daily over the entire experiment (93 d). Fragments were photographed using a Sony Handycam HDR-XR520, coupled to a Zeiss Stemi 2000-C stereomicroscope, on the first day of the experiment and also when the mouth and the complete polyp were first observed. A more thorough description of the whole regeneration process was achieved by examining two randomly chosen fragments three times a day (Supplementary Material 1 and 2). The area of tissue fragments was estimated at the beginning of the experiment and at the time of full polyp development using the software ImageJ 1.50b (http://imagej.nih.gov/ij/index.html).

2.2. Statistical analyses

2.2.1. Survival

Only 41 (17.1%) out of 240 fragments held in the laboratory underwent tissue necrosis and death. Because temperature is known to affect tissue metabolism in regenerating corals (e.g., Kramarsky-Winter and Loya, 2000; Denis et al., 2013; Bonesso et al., 2017; Burmester

et al., 2017) we tested whether survival rate differed between species and among temperature treatments (other factors pooled), using survival analysis. The Meier-Kaplan curve (Taylor et al., 1958) was applied to examine the correlation between experimental treatments and the percentage of surviving fragments over time. The Cox Proportional-Hazards Regression (Fox and Weisberg, 2011) was used to evaluate the significance of each treatment. This method assumes that the hazard of an event (e.g., tissue necrosis) at a given time can be modeled by the product of a baseline hazard function and a regression function of tested covariates (Sinnott and Cai, 2016). Pairwise comparisons (Peto and Peto, 1972) were undertaken to further examine significant covariates. Analyses were performed using the software R 3.2 applying the *survfit* and *coxph* from the Survival package (Therneau and Lumley, 2016).

2.2.2. Double-polyp regeneration

Twenty-one fragments of the whole sample (9%) underwent an alternative regeneration pattern, with the formation of two rather than one polyp. Although potentially increasing the chances of fragment development to healthy colonies, owing to fusion and increased growth thereafter (Forsman et al., 2015), the incidence of double-polyp regeneration could depend on size and metabolism of remnant tissues, and also energy resources. We tested this hypothesis by comparing the proportion of double-polyp regeneration between fragment size, temperatures and food supply conditions, for both *T. coccinea* and *T. tagusensis*, using two-tailed *Z*-tests.

2.2.3. Tissue retraction and time to regeneration

Percent retraction, as a proxy of the relative metabolic cost for tissue reorganization, and the time (in days) for the development of the mouth and tentacles (i.e., full polyp) were examined through separate 4-way analyses of variance. 'Species', "fragment size", "food supply" and 'temperature' were all considered fixed factors. The Cochran's *C* statistic was used to test data homoscedasticity and the SNK procedure was used for pairwise comparisons to further examine significant sources of variation. Balanced datasets were run in all these analyses, after random removals of entries exceeding the minimum sample size among all factor combinations [i.e., *T. tagusensis*; small fragment; food added; 27 °C ($n = 5$)]. The square-root transformation did not solve variance heterogeneity in the case of time-to-polyp data, owing to exceedingly high variation in a specific combination of factors (i.e., *T. coccinea*; large fragment; no food added; 24 °C). The procedure was, however, maintained, since heteroscedasticity does not prevent a safe interpretation of results in a balanced design with adequate sample size (Underwood, 1997).

3. Results

3.1. Survival and regeneration types

Regardless of species identity, the survival of coral fragments was only affected by temperature (food supply and fragment size did not affect survivorship – see Table 1 and Fig. 2). Survival rate was higher at 24 °C compared to 27 °C and 30 °C (Peto & Peto test, $p < 0.001$), with no differences between 27 °C and 30 °C ($p > 0.05$). Details of the regeneration patterns of surviving fragments, which developed to full polyps, are given in Fig. 3. Following initial tissue retraction, mouth rudiments, sometimes two for a single fragment, became noticeable. Subsequent development may either consist of tissue reorganization around mouth rudiments, leading to the formation of two small distinct polyps (Fig. 3a), or reabsorption of one of the rudiments. In the latter, major tissue differentiation around the remaining mouth rudiment were observed, resulting in a larger polyp (Fig. 3b). Temperature and food supply did not exert significant variance on the formation of number of polyps. However, fragment size apparently plays a role on the determination of development pathways in *T. tagusensis*. The proportion of surviving fragments developing 2-polyp individuals differed only between fragment size, being significantly higher in large compared to small fragments for *T. tagusensis* (0.24 vs 0; $z = 3.66$, $p < 0.001$), but not for *T. coccinea* (0.12 vs 0.06; $z = 0.92$, $p = 0.356$).

Table 1
Summary results for the Cox model testing fixed effects of species (*Tubastraea coccinea, T. tagusensis*) and temperature (24, 27, 30 °C) on the survival of coral fragments (41 mortality events on 240 subjects). Bold indicates that the p value is statistically significant.

Effect	*d.f.*	χ^2	*p*
Species: Sp	1	0.13	0.721
Temperature: Te	2	15.44	< **0.001**
Sp*Te	2	4.36	0.113

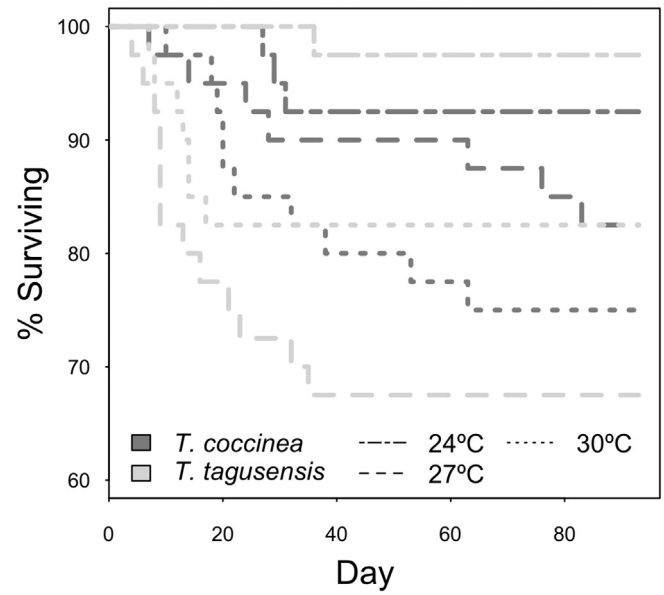

Fig. 2. Survival curves for *Tubastraea coccinea* and *T. tagusensis* coral fragments maintained in laboratory at different temperature regimes.

3.2. Tissue retraction and time to regeneration

Temperature exerted a pervasive and strong effect on all measured variables (Table 2). Tissue retraction, as a measure of overall reorganization, was higher at 24 °C and 30 °C compared to intermediate temperature (27 °C), with an effect size on the order of 40–70% (Fig. 4a). In spite of the significant 2-way interaction between 'fragment size' and 'food supply' (Table 2), post-hoc tests failed to confirm any difference between fed and unfed groups in each fragment size category. Therefore, the fragment size main-effect was retained, indicating larger percent retraction in small compared to large fragments (Fig. 4b).

Results generally support faster regeneration rates at higher temperatures (Table 2, Fig. 5), but some important interactive effects were identified. Fastest mouth regeneration at higher tested temperature was found for individuals without contact with food, with estimated times decreasing from around 23 d at 24 °C to 18 d at 30 °C. However, upon contact with living zooplankton, individuals held at 27 °C underwent much faster (~30%) mouth development compared to their counterparts (Fig. 5a). This explains the significant second-order interaction between food supply and temperature (Table 2), and suggests optimal mouth development at intermediate temperature, provided contact with potential food. However, this advantage is cancelled out when examining the development time to full polyps (tentacles added; Fig. 5b, c). Within a narrow range of variation, tissue fragments develop to complete polyps in c.a. 25 d at 27 °C and 30 °C, for both species. Besides the thermal effect, which delayed development to 30–32 d, food

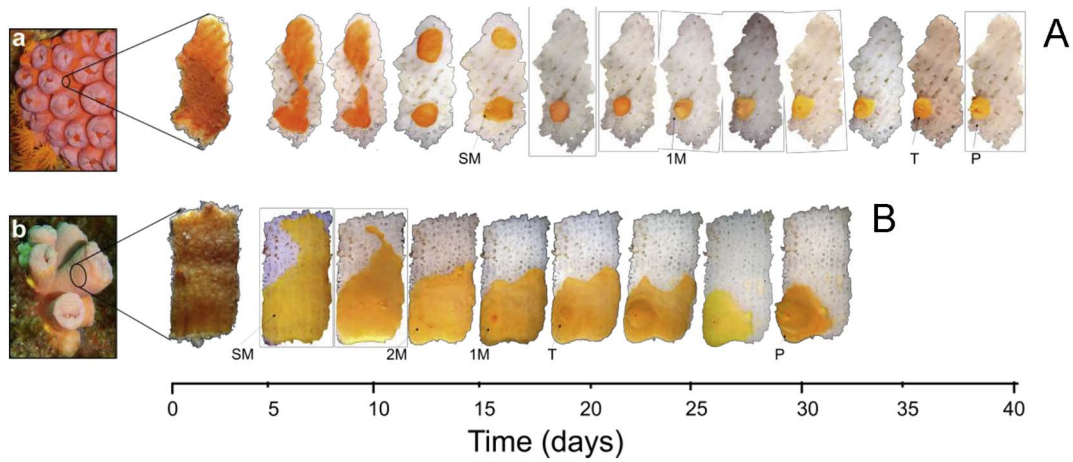

Fig. 3. Detailed regeneration process in large fed fragments of (A) *Tubastraea coccinea* (24 °C) and (B) *T. tagusensis* (30 °C), showing the formation of two mouth rudiments that may either (A) lead to the development of two separate polyps or (B) further reorganization with the loss of one of the openings and the development of a larger polyp. "SM": Starting mouth formation. "1 M" and "2 M": 1 or 2 mouth rudiments, respectively. "T": Tentacles. "P": Fully formed polyp. Scale bar: 5.0 mm.

provision was also critical for *T. coccinea* regeneration at 24 °C. Unfed individuals of this species took around 41d for polyp formation (Fig. 5b). This particular response of *T. coccinea* at the lowest tested temperature explains the third-order interaction presented in Table 2.

4. Discussion

This study reports surprising regeneration rates of sun corals *T. coccinea* and *T. tagusensis*, from undifferentiated tissue in small fragments completely detached from 'mother' colonies. Substantial tissue retraction, occasionally leading to discrete tissue patches (as in Fig. 3a), suggests that regeneration in *Tubastraea* is not direct, as observed in the aboral regeneration of *Hydractinia* (Bradshaw et al., 2015). Instead, remnant tissue from broken *Tubastraea* colonies are likely to be pluripotent, with ubiquitous spread of stem ectodermal, endodermal, and mesogleal cells, as in fungiid corals (Kramarsky-Winter and Loya, 1996; Vizel et al., 2011). Contrary to *Fungia granulosa*, however, which develops planula-like aggregates that become new anthocauli, the skeleton appears to be essential in the regeneration process for sun corals. Provided minimum skeleton area, i.e., 3.5 mm^2 as observed in this study, but possibly smaller, regeneration to viable polyps was shown to be the most likely outcome, rather than the exception.

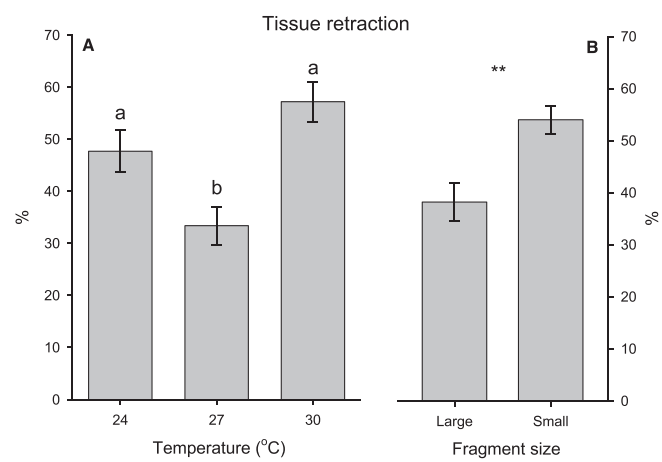

Fig. 4. Tissue retraction in regenerating polyp fragments according to (A) temperature and (B) fragment size. Bars and whiskers stand for average and respective ± 1SE estimates. In (A) different letters above bars indicate statistical significant differences ($p < 0.05$). In (B) ** indicates significance at $\alpha = 0.01$.

Table 2
Summary results of 4-way analyses of variance testing the effects of species (*Tubastraea coccinea, T. tagusensis*), fragment size (small, large), food supply (low, high) and temperature (24, 27, 30 °C) on the extension of tissue retraction (%) and time (d) for the development of the mouth and tentacles (i.e., full polyp development). Significant sources of variation are highlighted in bold ($p < 0.05$).

		% Retraction			Time to mouth			Time to polyp		
	d.f.	MS	*F*	*p*	MS	*F*	*p*	MS	*F*	*p*
Species: Sp	1	526.9	1.01	0.317	18.4	0.43	0.512	< 0.001	< 0.01	0.986
Fragment size: Fr	1	7522.7	14.44	**< 0.001**	46.9	1.10	0.296	0.137	0.30	0.586
Food supply: Fo	1	15.6	0.03	0.863	15.4	0.36	0.549	0.144	3.15	0.079
Temperature: Te	2	11,551.7	11.09	**< 0.0001**	239.4	5.63	**0.005**	5.749	12.56	**< 0.001**
Sp*Fr	1	24.6	0.05	0.829	85	2.00	0.160	0.021	0.05	0.829
Sp*Fo	1	68.3	0.13	0.718	52	1.22	0.271	0.319	0.70	0.451
Sp*Te	2	1082.5	1.04	0.358	19.9	0.47	0.628	0.995	2.17	0.119
Fr*Fo	1	3082.0	5.92	**0.017**	54.7	1.29	0.260	< 0.001	< 0.01	0.983
Fr*Te	2	464.8	0.45	0.641	27.1	0.64	0.531	0.286	0.63	0.534
Fo*Te	2	2862.8	2.75	0.069	141.5	3.33	**0.040**	0.085	0.18	0.831
Sp*Fr*Fo	1	26.6	0.05	0.822	5.2	0.12	0.727	0.445	0.97	0.327
Sp*Fr*Te	2	1269.7	1.22	0.300	23.5	0.55	0.577	0.449	0.98	0.379
Sp*Fo*Te	2	141.6	0.14	0.873	57.6	1.35	0.263	1.879	4.10	**0.020**
Fr*Fo*Te	2	447.8	0.43	0.652	27.5	0.65	0.526	0.070	0.15	0.858
Sp*Fr*Fo*Te	2	559.3	0.54	0.586	73.4	1.73	0.184	0.450	0.98	0.378
Error	96	520.8			42.5			0.4577		
		$C = 0.0942$; $p > 0.05$			$C = 0.1333$; $p > 0.05$			$C = 0.3047$; $p < 0.01$		
		Raw data			Raw data			$\sqrt{(X + 1)}$ transformed		

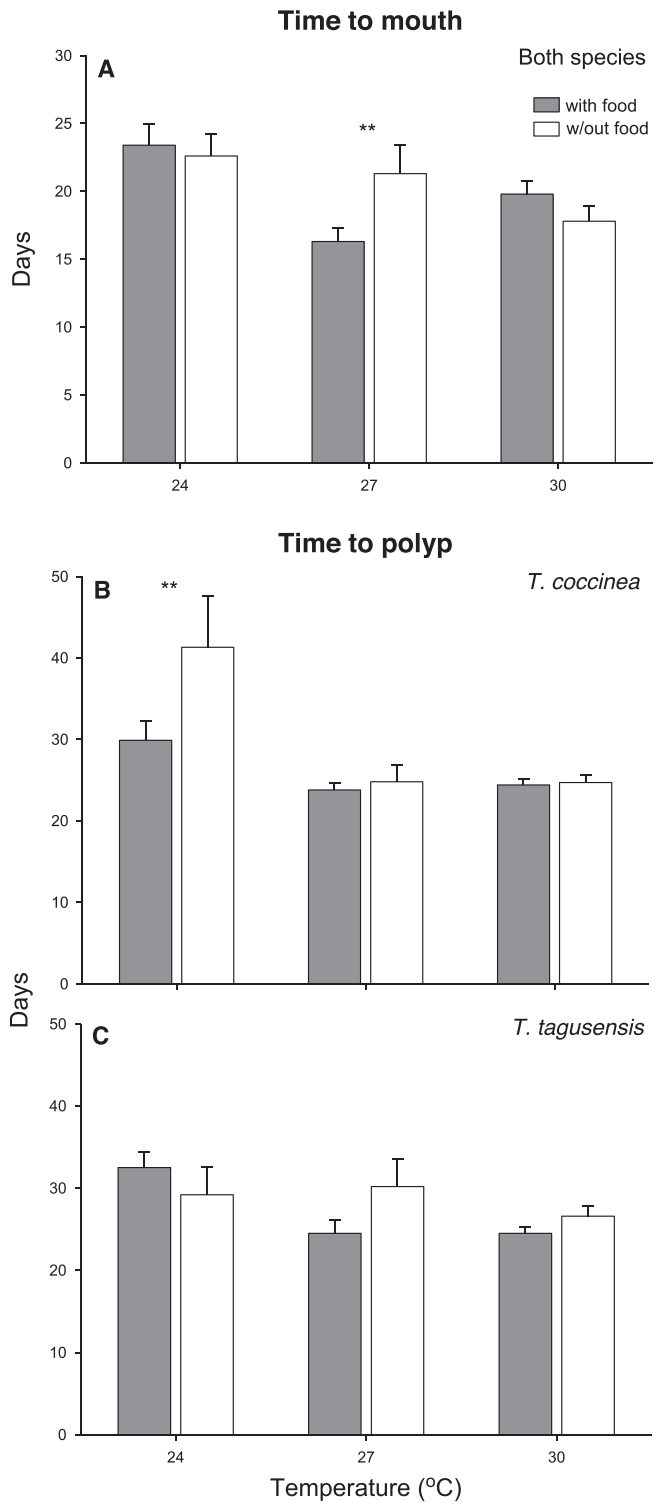

Fig. 5. Time for the development of (A) mouth and (B, C) tentacles (full polyp). Plots show the 2-way interactive effects of food supply and temperature in the formation of mouth (A) and the 3-way interactive effect, with the addition of species identity, in the formation of full polyps (B, C). Bars and whiskers as in Fig. 4. $^{**}p < 0.01$.

Overall survival at the end of the experiment (93 d) was high (83%), especially at the lowest tested temperature (24 °C), at which survival reached 95%. This suggests that the largest deposits of living coral fragments observed at Búzios Island would likely build up over winter, when cold fronts generating sufficient wave action to impact sun coral colonies are more frequent (Dottori and Castro, 2009), and temperature conditions favorable for highest survival of coral fragments.

Percent tissue retraction, as a measure of metabolic cost, was lowest at the intermediate tested temperature, which is close to the average year-round temperatures in the tropical regions from which those species are native (Cairns, 1991; Fenner and Banks, 2004), but typical of average summer surface temperature at the subtropical study area. There is no evidence of direct negative effects of increased metabolic cost, since the highest average survival was observed at 24 °C (but had the highest tissue retraction), not 27 °C. There might be, however, important indirect effects as tissue retraction is inversely related to polyp size; a critical variable determining early mortality in natural conditions, since small coral recruits (or regenerating fragments) are more vulnerable to environmental disturbance such as incidental fish grazing (Doropoulos et al., 2009) or sediment smothering (Wittenberg and Hunte, 1992). Fragment size was found to exert an additive effect to temperature in the determination of tissue retraction. Contrary to predictions of tissue reorganization to a standard polyp size, the percent retraction in small fragments was larger (not smaller) than in large fragments. This demonstrates that polyps regenerating from small fragments are metabolically more demanding and also disproportionally smaller. It is interesting to note, however, that despite the disadvantages of small polyp size, most cases of two-polyp regeneration, leading to much smaller individual size (Fig. 3) were observed in some large fragments. As observed in other scleractinians (Forsman et al., 2015), it is possible that tissue spreading and fusing later in development, may actually render higher growth rates and faster substrate cover, a possibility that deserves further investigation.

In accordance with temperature-dependent tissue retraction, the dynamics of mouth formation further indicated that the regeneration process is favored at intermediate temperature. Fast average mouth development in just 16d at 27 °C is nevertheless dependent on the presence of food, which likely acts as a trigger. In the laboratory, any specific advantages at 27 °C cancelled out later on, with faster development to full polyps at both 27 °C and 30 °C. It is important to note that native zooxanthellate scleractinian species along the Brazilian coast extensively bleach one week after exposure to 29–31 °C, corresponding to anomalous sea surface temperatures of 1–1.7 °C compared to historical data (e.g., Migotto, 1997; Ferreira et al., 2013; Miranda et al., 2013). As such, sun corals may undergo successful regeneration at thermal conditions that impair potential coral competitors. While otherwise exhibiting remarkably similar regeneration patterns, *T. coccinea* may take longer than *T. tagusensis* for complete polyp development when fragment is kept unfed at 24 °C.

Thus, regeneration of small skeleton fragments to new viable polyps may be an important means of asexual reproduction and small-scale dispersal in both *T. coccinea* and *T. tagusensis*. Besides wave action, possible fish-coral interactions (Mizrahi et al., 2016), may cause coral breakage/fragment dislodging, and thus contribute to the pool of regenerating coral fragments. Thus, although no doubts remains on the necessity and efficacy (de Paula et al., 2017) of invasive *Tubastraea* spp. management efforts, the results presented herein, including those related to temperature, should be taken into account during such processes. Also, a better understanding of the fate of fragments in the wild, i.e., the likelihood of being trapped and reattached to local reef habitat, or thrive as small polyps in adjacent soft-bottom habitats, becomes an important question for the management and eradication of invasive sun coral populations.

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jembe.2018.02.002.

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